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Some thesis title

by

Mostert

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Supervisor: Prof. Name Surname

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Someone.

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- Supervisor X.
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“ *A very long quote*
from someone very important. ”
– Mr M

Thank you to Leanne for helping to prepare this template.

Declaration

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Abstract

Uittreksel

Table of Contents

Abstract iii

Uittreksel iv

Table of contents v

List of figures vii

List of tables vii

Nomenclature ix

Acronyms and abbreviations x

1. Introduction 1

1.1. Background 1

1.2. Problem Statement 2

1.3. Objectives 2

1.4. Summary of Work 2

1.5. Scope 2

1.6. Roadmap 3

2. Literature 4

2.1. Related Work 4

3. System Methods 5

3.1. Dataset 5

3.1.1. Annotations 5

3.1.2. Facility Structure 7
3.1.3. Data Preparation 7
3.1.3.1. Excluded from the dataset 9

4. Summary and conclusions **11**

References **12**

A. Appendix title goes here **13**

List of Figures

List of Tables

3.1. Summary of the dataset by facility 7
3.2. Aerobic zone crops per class 9
3.3. Clarifier crops per class 9

Nomenclature

Acronyms and Abbreviations

MFT My Funky Thesis

Variables and Functions

Chapter 1

Introduction

1.1 Background

Municipal wastewater treatment works (WWTW) are made up of several process units that work together in sequence. Larger plants usually rely on the activated sludge process, which uses biological reactors full of suspended solids (activated sludge) plus secondary settling tanks called clarifiers.

Inside the biological reactors, treatment happens in unaerated and aerobic zones. The aerobic zones need mechanical aeration to push oxygen into the water so bacteria can break down waste. There are several ways to do this, but Fine Bubble Diffused Aeration (FBDA) is one of the most common. Once the water has been aerated, it moves into circular clarifiers, which separate clear effluent from the sludge. Gravity pulls the sludge down into a central hopper, where it gets sent back to the bioreactor to treat more incoming wastewater.

Interestingly, someone who knows what to look for can tell a lot about how these systems are performing just from aerial or satellite photos. A healthy FBDA unit churns the surface into a "leopard print" pattern, while a struggling one looks noticeably smoother. Clarifiers tell a similar story: when they're working well, they separate out the solids and leave behind a clear liquid that looks black from above. When they're not working, solids escape into the effluent and the surface turns green, cloudy, or uneven.

1.2 Problem Statement

Right now, checking on how well WWTW units are running mostly comes down to on-site sensors, physical inspections, or someone experienced reading aerial photos by eye. That's slow, and it doesn't scale well once you're trying to keep tabs on multiple facilities at once. The question this project asks is simple: can a computer vision system pick up on the same visual cues a trained person would (the aerator's "leopard print," the clarifier's colour) and use them to judge performance? This project tests whether historical satellite imagery can be used to automatically monitor how well FBDA aerators and clarifiers are functioning.

1.3 Objectives

To answer that question, the project sets out to do three things:

1. Prepare the data and extract features by working with a set of labelled Google Earth satellite images taken on different dates to capture the visual states of WWTW components.
2. Classify the states using a CNN by training a convolutional neural network to sort Aerobic Zones into three categories (Functional, Sub-optimal, Dysfunctional) and Clarifiers into four categories (Functional, Dysfunctional, Scum, Empty), based on how their surfaces look.
3. Test whether it actually works ....

1.4 Summary of Work

1.5 Scope

Included: Fine Bubble Diffused Aeration (FBDA) in aerobic zones, and secondary clarifiers. The project only looks at the Atlantis, Cape Flats, and Waterval WWTW facilities.

Excluded: Surface-mounted aerators and anaerobic zones aren't covered. There's no live satellite feed, no physical water sampling, and no hardware devices deployed on site — everything is based on the historical Google Earth imagery already available.

1.6 Roadmap

- Chapter 2 (Literature Review): Looks at existing monitoring solutions for wastewater management, remote sensing tech, and the computer vision concepts this project builds on.
- Chapter 3 (System Methods): Covers the system's architecture, how the dataset is structured, and the metrics used to judge the model's performance.
- Chapter 4 (Detailed Design): Gets into the technical details of the image processing pipeline and the classification logic.
- Chapter 5 (Results): Presents the data, the performance numbers, and what they say about how accurately the system classifies each unit.
- Chapter 6 (Conclusion): Wraps up the findings, checks how well the objectives were met, and suggests where future versions of this kind of monitoring could go.

Chapter 2

Literature

2.1 Related Work

Chapter 3

System Methods

3.1 Dataset

The dataset used in this project consists of satellite imagery sourced from Google Earth, covering three WWTW facilities: Atlantis, Cape Flats, and Waterval. Each image is a snapshot of a facility's biological reactors at a particular point in time, with filenames encoding the capture date in a `month_year` format.

Each biological reactor in the dataset consists of an aerobic zone and one or more clarifiers, and these are the two components the project focuses on classifying.

3.1.1 Annotations

Images were manually annotated using the VGG Image Annotator (VIA2) tool. Aerobic zones were labelled using rectangular bounding boxes, while clarifiers were labelled using circular regions, reflecting the physical shape of each structure. Each annotation carries a region attribute indicating the visible state of that component at the time the image was captured.

Aerobic zones were labelled into one of three categories, based on how much of the surface exhibits the characteristic "leopard print" aeration pattern:

- **Functional**: the surface shows the leopard print pattern.
- **Suboptimal**: a third or more of the surface shows no leopard print, or the state is unclear.
- **Dysfunctional**: two thirds or more of the surface shows no leopard print, or is clearly dysfunctional.

Clarifiers were labelled into one of four categories, based on surface colour and visibility:

- **Functional**: surface appears uniformly black.
- **Dysfunctional**: surface appears green.
- **Scum**: surface is not uniformly black.
- **Empty**: shadows are visible inside the tank.

Annotations are stored as VIA2 project files in JSON format. Each entry records the image filename, file size, and a list of regions, where each region specifies its shape (rectangle or circle), pixel coordinates, and the associated class label. An example entry for an aerobic zone annotation is shown below:

```
"Atlantis 01_2019.jpg3469910": {
  "filename": "Atlantis 01_2019.jpg",
  "size": 3469910,
  "regions": [
    {
      "shape_attributes": {
        "name": "rect",
        "x": 2707, "y": 931,
        "width": 2839, "height": 1381
      },
      "region_attributes": { "aerobic zone": "Dysfunctional" }
    }
  ],
```

```
  "file_attributes": {}
}
```

3.1.2 Facility Structure

The three facilities differ slightly in how their data is organised, reflecting differences in plant layout.

Atlantis consists of a single biological reactor, so its 48 images are stored in one folder with no subdivisions. Aerobic zone and clarifier labels are each stored in a single JSON file.

Cape Flats consists of four separate reactor units (CF1–CF4), each with its own image folder and containing between 92 and 100 images. Each unit has its own aerobic zone and clarifier label file.

Waterval consists of two reactor units, referred to as Top and Bottom, with 37 and 36 images respectively. Each unit has its own aerobic zone and clarifier label file.

Table 3.1 summarises the structure of the dataset across the three facilities.

Table 3.1: Summary of the dataset by facility

Facility	Units	Images
Atlantis	1	48
Cape Flats	4	382 (96+94+92+100)
Waterval	2	73 (37+36)

This structural difference between facilities — a single unit at Atlantis, four at Cape Flats, and two at Waterval — is factored into the data pipeline described in Section 3.1.3, since each facility's images and labels need to be parsed slightly differently before they can be fed into the CNN.

3.1.3 Data Preparation

Before the labelled regions can be used to train a CNN, they need to be extracted from the full facility images and organised by class. This is handled by a preparation script (`prep.py`), which reads the VIA2 annotation files, crops out each labelled region, and saves the crops into a folder structure that can be loaded directly with `torchvision.datasets.ImageFolder`.

The script processes each facility one unit at a time, since Atlantis, Cape Flats, and Waterval each have a different number of units and label files. For every unit, it opens the aerobic zone and clarifier JSON files separately and works through each image entry in turn.

For each labelled region in an image, the script performs the following steps:

- Reads the shape attributes of the region. Aerobic zones are stored as rectangles, defined by an x and y coordinate along with a width and height. Clarifiers are stored as circles, defined by a centre point and a radius.
- Converts the shape into a bounding box that can be used to crop the image, since Python's Image Library crop function requires a rectangular box even for the circular clarifier regions.
- Clamps the box to the edges of the image, in case a region was drawn slightly outside the image bounds during annotation.
- Reads the class label from the region attributes and checks it against the expected set of classes for that component. Labels are also normalised, so that small inconsistencies in capitalisation from the manual annotation process do not create extra classes.
- Crops the region out of the source image and saves it as a JPG file inside the folder matching its class label.

Each output filename encodes the facility, unit, and original image name, so that a crop can always be traced back to the source image it came from. For example, a crop from unit CF2 at Cape Flats would be saved with a filename beginning `CapeFlats_Images-CF2_...`, followed by the original image name and a region number.

The result is a folder structure with one top-level folder per component and one subfolder per class:

```
cnn_dataset/
    aerobic_zone/
        Functional/
        Suboptimal/
        Dysfunctional/
    clarifier/
        Functional/
        Dysfunctional/
        Scum/
        Empty/
```

This structure means the aerobic zone and clarifier crops are treated as two separate classification problems, each with its own set of class folders, rather than a single combined dataset.

Table 3.2 and Table 3.3 summarise the number of crops produced for each class once the full dataset was processed.

Table 3.2: Aerobic zone crops per class

Class	Crops
Functional	693
Suboptimal	223
Dysfunctional	805

Table 3.3: Clarifier crops per class

Class	Crops
Functional	827
Dysfunctional	229
Scum	679
Empty	115

3.1.3.1 Excluded from the dataset

During processing, two annotated classes were found to have too few examples to be usable for training. The aerobic zone dataset contained a single region labelled **Empty**, and the clarifier dataset contained four regions labelled

Stagnant. Both classes were therefore excluded from the final dataset, and the corresponding regions were not cropped or saved. A class with only one or a handful of examples cannot be meaningfully split into training, validation, and test sets, and would not give the CNN enough examples to learn a reliable representation of that class. Including such a small class also risks skewing the overall class balance and encouraging the model to simply ignore it during training, since the loss contribution from so few examples is negligible compared to the other classes. The excluded regions were still counted separately by the preparation script, so their presence in the original annotations is not lost, but they play no further part in the classification task.

Chapter 4

Summary and Conclusions

References

[1] Stellenbosch University, "Formatting your thesis," 2024, http://bit.ly/SUThesisFormat.

[2] D. Els, "Stellenbosch university thesis bundle," 2024, https://ctan.org/pkg/stellenbosch-2.

Appendix A

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